1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In hepatocellular carcinoma (HCC), high PFKFB3-driven glycolytic reprogramming promotes cancer stem cell–like properties (Stem.Sig) preferentially in tumors with low stromal-to-tumor ratio (low TSR). This effect is potentiated by EPB41L2-associated cytoskeletal/adhesion remodeling that reconfigures the tumor microenvironment to support stem-like cells. Consequently, HCC tumors that simultaneously meet: (i) low TSR, (ii) high PFKFB3 expression, and (iii) a high EPB41L2 risk signature will exhibit elevated Stem.Sig activity and poorer overall survival (OS) than tumors lacking this metabolic–stromal interplay. The interaction is at least partly mediated through mTOR–linked growth and adhesion/ECM remodeling pathways, suggesting that disrupting PFKFB3-driven glycolysis or EPB41L2-dependent adhesion could mitigate stemness and improve prognosis in this subset.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PFKFB3, EPB41L2
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PFKFB3: PFKFB3-linked glycolytic programs are tied to cancer stemness and poor prognosis in HCC, yet stemness is reported to be inversely related to stromal content — so why do some tumors with low stroma and high glycolysis become clinically aggressive while others do not? 2) Tier 1 paths used: (a) PFKFB3 → regulates → Cancer stemness (Stem.

EPB41L2: EPB41L2 → observed as → underlying risk factor (provides a genetic/adhesion-related context that could modulate microenvironmental niche and stemness outcomes).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Cancer stemness (Stem.Sig)	Supporting modifier from knowledge graph

Secondary	stroma content	Supporting modifier from knowledge graph
Secondary	mTOR	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: drives stemness (Stem.Sig), modulated by the microenvironment (EPB41L2 and TSR) and amplified via mTOR signaling. This yields a testable prognostic frame: in HCC, the combination of low TSR, high PFKFB3, and elevated EPB41L2 risk signature should prognosticate worse OS due to enhanced stemness and niche support.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- Cancer stemness (Stem.Sig): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- stroma content: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.